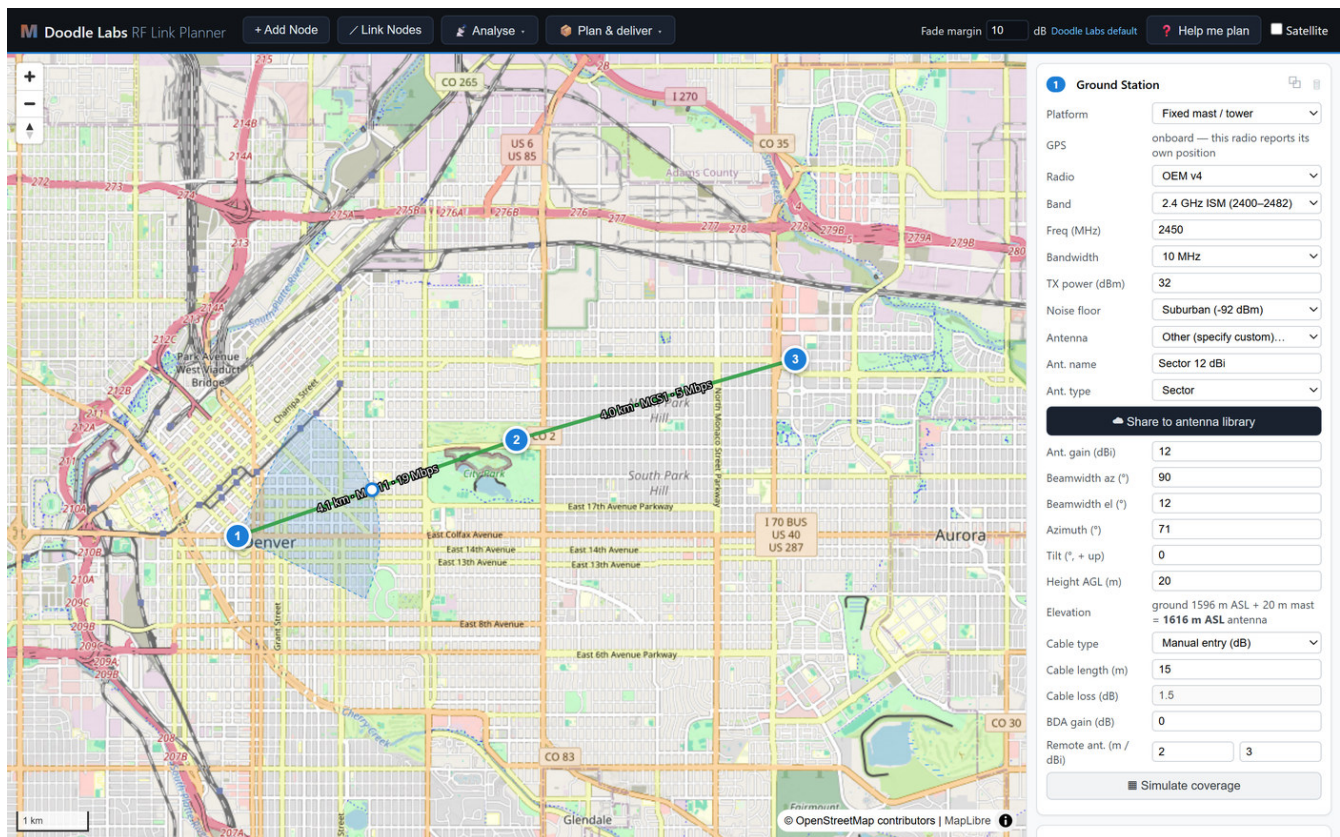


Using the Link Planner

Designing Mesh Rider networks on real terrain - a practical guide



Written for anyone who has to answer "**will this link work?**" - whether or not you design radio networks for a living. No RF background is assumed. Terms in **bold** the first time they appear are explained in [Definitions](#) at the back.

Doodle Labs Solutions Engineering · <https://doodlesim.jus2419497.workers.dev>

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1. What this tool is

The Link Planner answers one question: **if I put radios here, will they talk to each other, and how well?** It does that against real ground - actual hills and valleys from elevation data - rather than assuming the world is flat.

It will tell you:

- Whether two radios can see each other over the terrain between them
- What data rate to expect, and how much signal you have spare
- Where a whole network reaches, drawn as a heat map on the map
- How low a drone can fly and stay connected
- Whether the link can carry your video and control traffic, not just connect
- What to buy and where to put it, if you would rather describe the job than design it

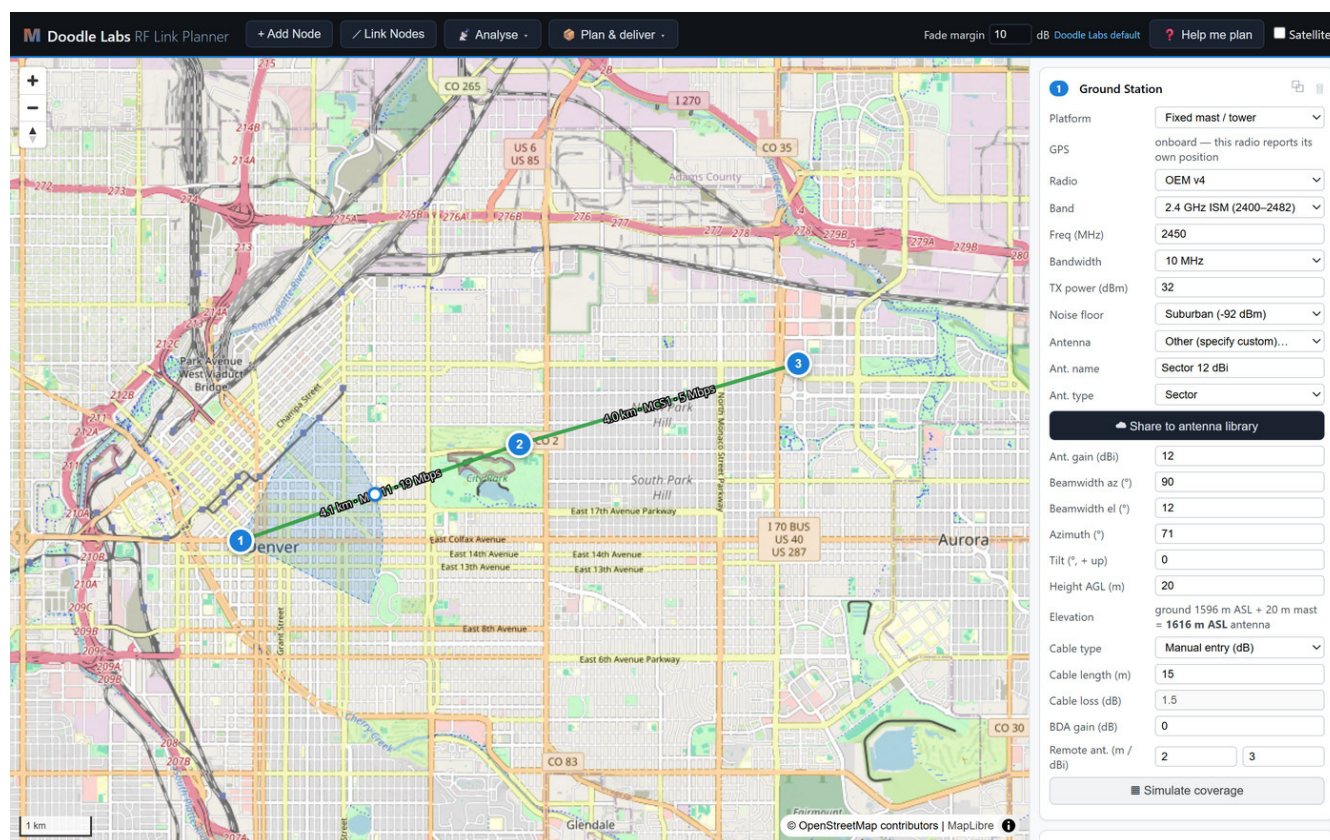
What it does not know

Buildings, trees and vehicles are **not** in the terrain data - only the shape of the ground. In a city or a forest, treat the answer as the best case and add margin. It also cannot know about interference that is not there yet.

2. The five-minute version

If you only read one page, read this one.

1. **Place your radios.** Add Node, then click the map where each one will be.
2. **Set each one up.** Pick the radio model, band, antenna and how high it is mounted. Height matters more than almost anything else.
3. **Join them.** Link Nodes, then click two radios. The line turns green, amber or red.
4. **Look at the result.** The card tells you the distance, whether the path is clear, the expected data rate and the spare signal.
5. **Check coverage** if you need to know where else you could go, and the **link budget** if you need to know whether your video will fit.
6. **Export a report** when you are happy, or save the layout to come back to.



The whole tool. Map and radios on the left, the selected radio's settings and the link results on the right.

3. Step 1 - Put your radios on the map

Click **+ Add Node**, then click the map. Each radio appears as a numbered circle you can drag at any time - every calculation updates as it moves, so it is worth dragging a mast around to see how much the answer improves.

- Place radios where they will *actually* be installed, not where it is tidy. A hundred metres along a ridge can be the difference between working and not.
- Use the **Satellite** switch, top right, to see what is really on the ground.
- Finished setting one up? Use the **duplicate** button on its card to copy it complete with every setting, then drag the copy where it belongs.

4. Step 2 - Set each radio up

Selecting a radio opens its settings. There are a lot of them, but only a few change the answer much.

The ones that matter most

- **Height above ground (AGL)** - the single biggest lever. Raising a mast usually beats buying a better antenna, because it clears obstructions rather than shouting through them.
- **Antenna and its gain (dBi)** - more gain reaches further but over a narrower beam, so a high-gain antenna must be aimed. Omnidirectional antennas cover everything and reach less far.
- **Band and channel width** - narrower channels reach further and carry less. If a link is marginal, narrowing from 20 MHz to 10 MHz is often the cheapest fix.

- **Noise floor** - how electrically busy the site is. A quiet field and a factory floor are not the same radio problem, and this is where you say so.
- **Cable type and length** - long thin coax quietly throws away signal before it reaches the antenna. The planner works the loss out from the type and run.

1 Ground Station

Platform	Fixed mast / tower
GPS	onboard — this radio reports its own position
Radio	OEM v4
Band	2.4 GHz ISM (2400–2482)
Freq (MHz)	2450
Bandwidth	10 MHz
TX power (dBm)	32
Noise floor	Suburban (-92 dBm)
Antenna	Other (specify custom)...
Ant. name	Sector 12 dBi
Ant. type	Sector

Share to antenna library

Ant. gain (dBi)	12
Beamwidth az (°)	90
Beamwidth el (°)	12
Azimuth (°)	71
Tilt (°, + up)	0
Height AGL (m)	20
Elevation	ground 1596 m ASL + 20 m mast = 1616 m ASL antenna

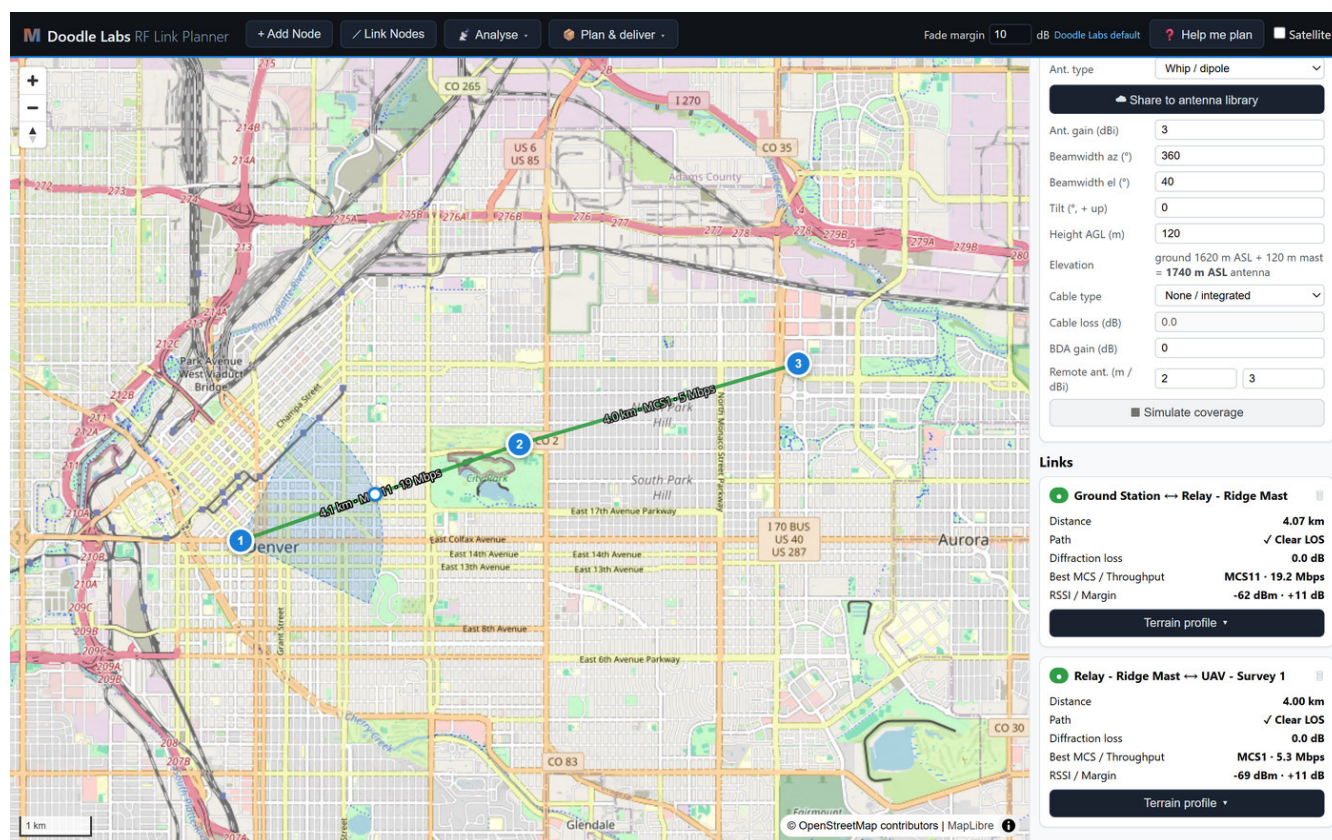
One radio's settings. Everything here affects the prediction, but the highlighted few dominate it.

If you set one thing carefully, set the height

Antenna height is the setting people guess at and the one the maths is most sensitive to. A mast height entered as a hopeful number rather than a measured one will make every other figure on the page wrong in the same direction.

5. Step 3 - Join them, and read the answer

Click **Link Nodes**, then click two radios. The planner cuts a slice through the elevation data between them and works out what the signal has to get past.



A finished link. The green line is healthy; the label shows distance and expected rate, and the card on the right gives the detail.

What the colours mean

- **Green** - comfortable. Signal to spare above your fade margin.
- **Amber** - works, but with little in hand. Fine for a demonstration, risky for a deployment.
- **Red** - will not carry a usable link as configured.

What the card tells you

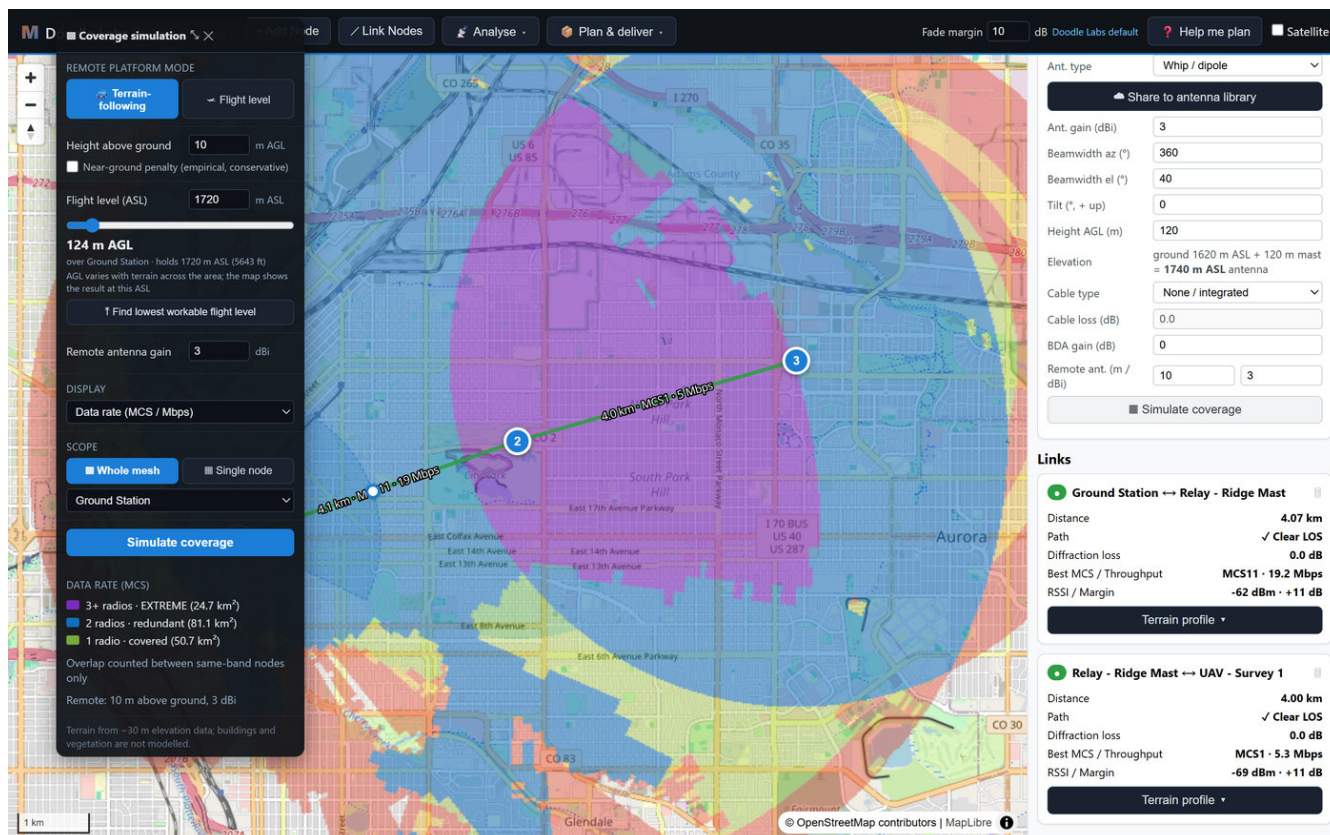
- **Distance** - straight-line distance between the two radios.
- **Path** - whether the ground gets in the way. "Clear LOS" means the **Fresnel zone** is unobstructed, not merely that you could see the far end.
- **Diffraction loss** - signal lost bending over whatever is in the way. Zero is what you want.
- **Best MCS / Throughput** - the data rate step the radio should settle at, and roughly what that delivers.
- **RSSI / Margin** - what the far radio hears, and how much spare signal there is above the minimum *after* your fade margin is taken out.

A red link is information, not a failure

Most useful designs start red. Raise a mast, narrow the channel, add gain, or put a relay on high ground between the two - then watch the same numbers move. That iteration is what the tool is for.

6. Step 4 - See where you can actually talk

A link tells you about two points. **Coverage simulation** paints every direction at once, so you can see the shape of what you have built.



Coverage across the whole network. Colour shows where one, two, or three and more radios reach - the darker areas have redundancy if one goes down.

Two ways to ask the question

- **Terrain-following** - for anything that stays near the ground: vehicles, handhelds, boats, people. Set a height above ground and the simulation follows the landscape up and down.
- **Flight level** - for aircraft, which hold an altitude rather than following the ground. Enter the altitude the aircraft holds and the readout tells you what that means in **AGL** over your reference point.

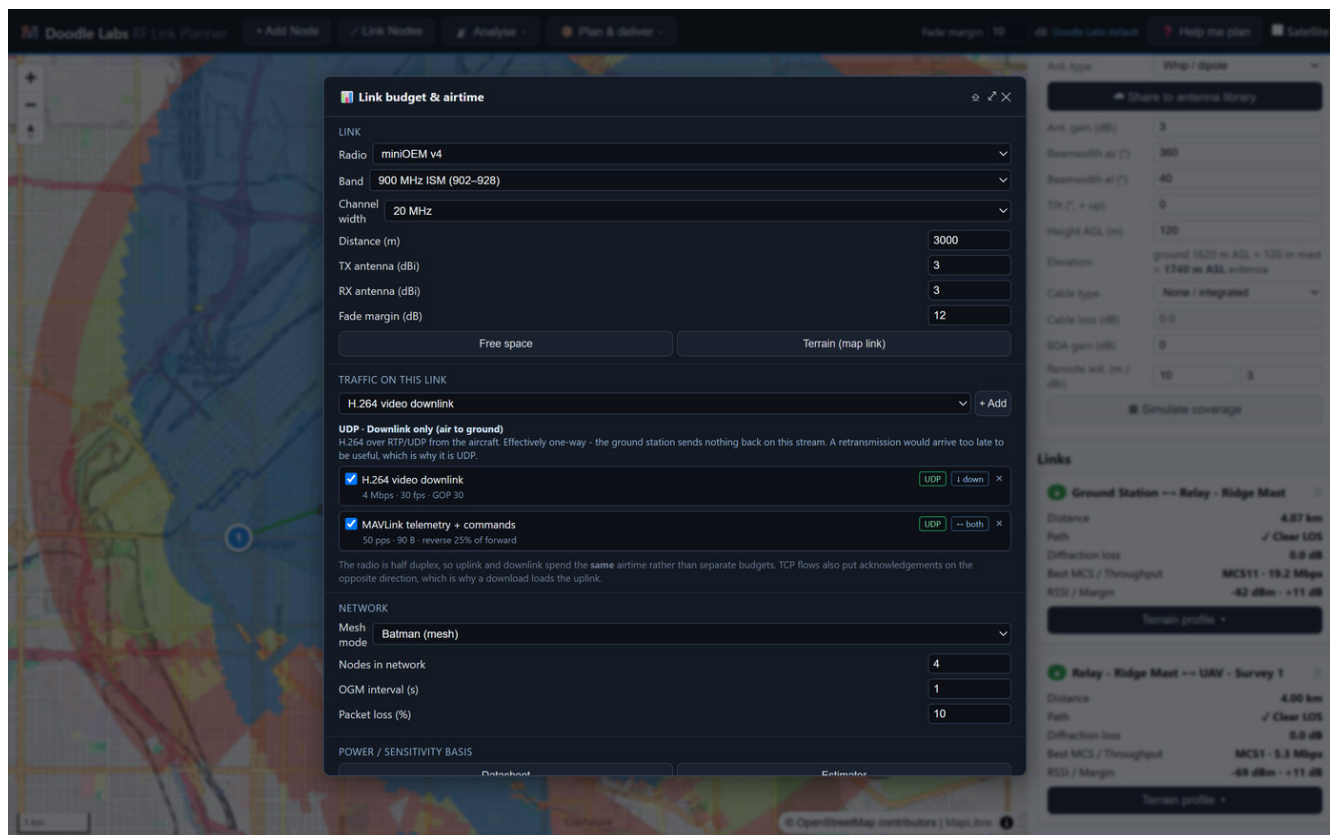
Flight level and height above ground are not the same number

An aircraft holding a constant altitude is getting closer to the ground as the terrain rises. The panel leads with the AGL figure because that is what pilots and regulations use, and shows the altitude actually held beneath it - but that AGL is only true over the reference point.

The **Find lowest workable flight level** button answers the question drone operators actually ask: how low can I fly and stay connected? It sweeps altitudes and stops at the lowest one that still covers the area.

7. Step 5 - Check it can carry your traffic

A link closing and a link doing its job are different questions. A radio can show a strong signal and still deliver stuttering video, because the channel is full. This is the panel that tells you.



The link budget and airtime panel. Add the traffic the link really carries and it reports how much of the channel each stream eats.

Adding your traffic

Pick from the list - H.264 video down from the aircraft, MAVLink telemetry both ways, an RC control uplink, a file transfer - and add as many as the link carries. Each one already knows whether it runs on **TCP** or **UDP** and which direction it flows, because that changes the answer.

Two things that surprise people

Uplink and downlink share one budget. The radio talks in turns on a single channel, so a 4 Mbps video feed down and a small control stream up are not separate allowances - they add up.

A download loads the uplink. A TCP transfer sends an acknowledgement back for roughly every two packets. Those are tiny messages but they cost full airtime, so a one-way-looking file copy is not one-way on the air.

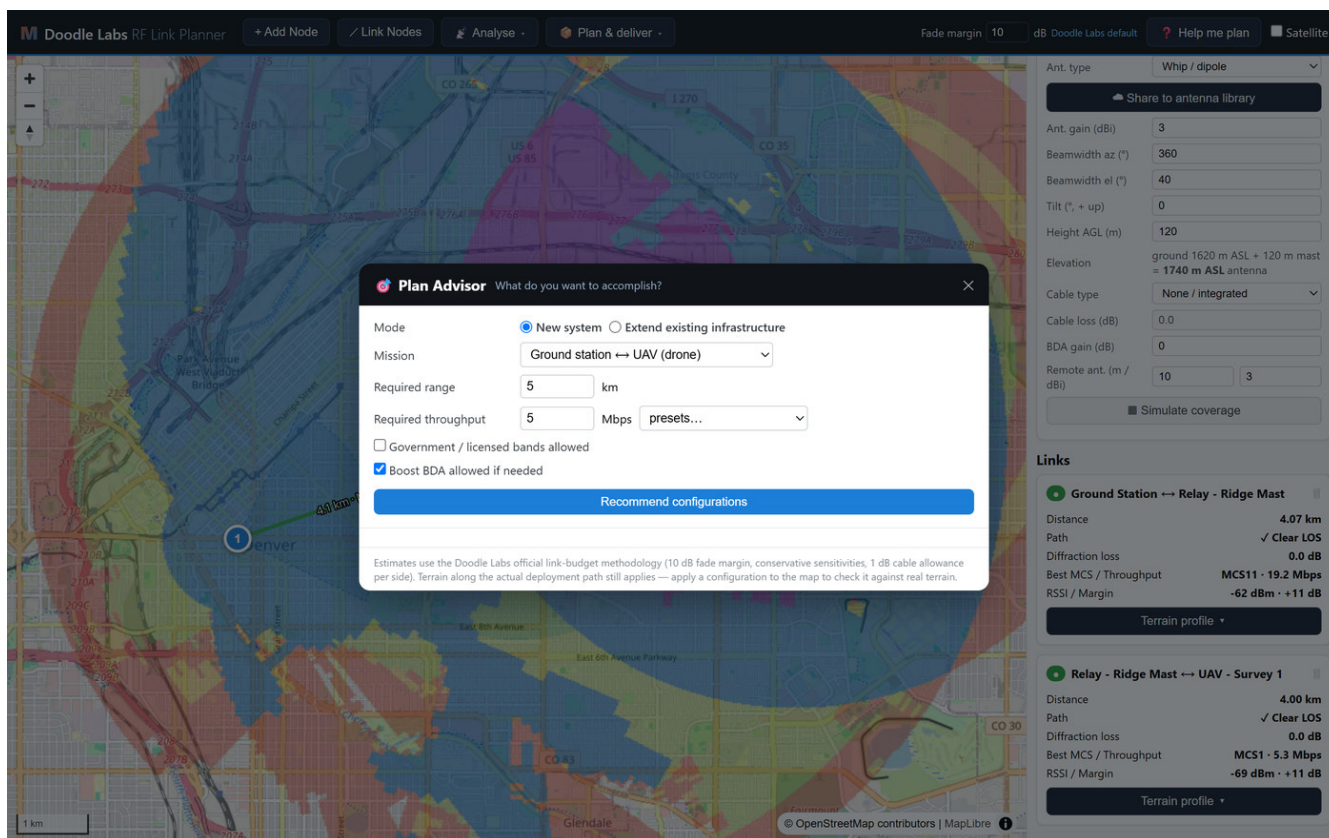
Reading the result

- **Under 50%** - comfortable.
- **50-75%** - busy. It works until something else needs the channel.

- **Over 75%** - saturated. Expect latency and dropped frames.
- **Peak above 100%** - the burst does not fit. Video sends a large keyframe about once a second, and if that cannot clear in time you get stutter even though the average looks fine.

8. Letting the Plan Advisor design it

If you would rather describe the job than design it, the **Plan Advisor** takes the mission - what needs to talk to what, how far apart, how much data - and returns ranked configurations: radio, band, antenna, aiming, mast height and cabling. It will work with radios you have already placed, and can site a relay when terrain blocks a single hop.



Describe the outcome; get back configurations that achieve it, each with the reasoning attached.

9. Saving work and handing it over

- **Save layout** writes a file you can email or archive. The browser also keeps your current work automatically, so closing the tab does not lose it.
- **Load layout** opens a saved design - useful for keeping a library of standard deployments to start from.
- **Report** exports a customer-ready document with the terrain profiles, link results and minimum drone altitudes worked out.

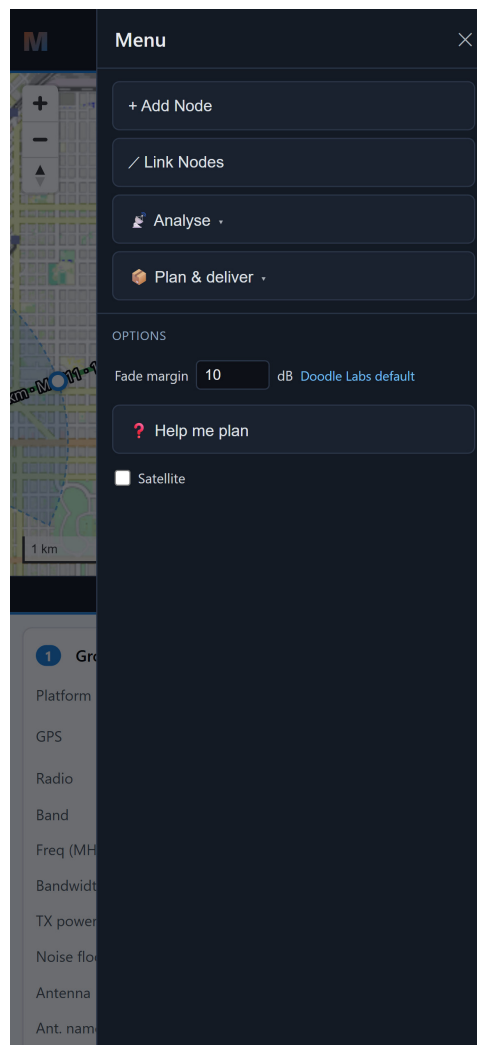
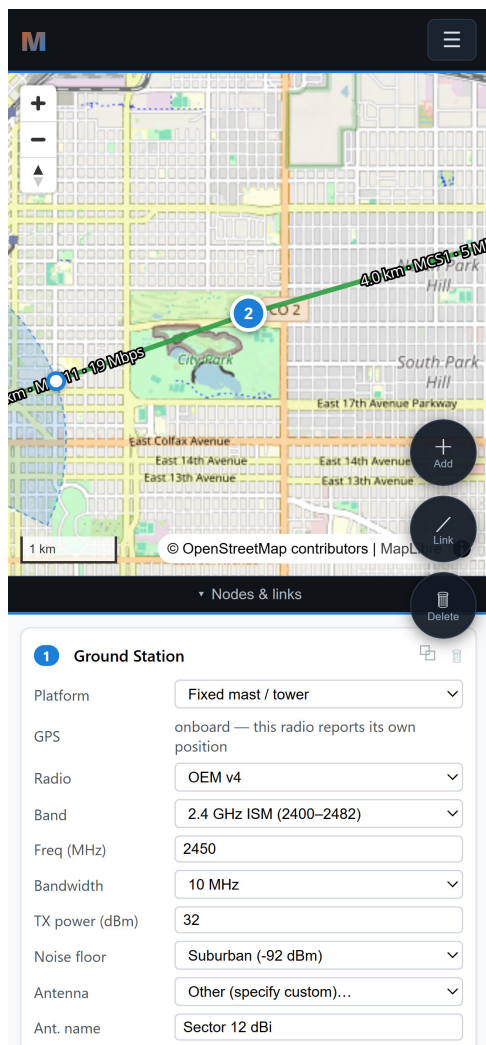
Send the layout, not just the picture

A screenshot shows what you concluded. The layout file lets the person receiving it change a mast height and see the consequence for themselves, which is usually the conversation you actually want to have.

10. Using it on a phone

The planner works on a phone, which matters when the question comes up on site rather than at a desk.

- The three round buttons on the map add a radio, join two, or delete one. Delete asks you to arm it first, then tap the radio - so a stray tap cannot remove anything.
- The **Nodes & links** handle collapses the list when you want the whole screen for the map.
- Panels slide up from the bottom instead of sitting in a corner.



Left: the map takes the screen, with the radio list collapsible beneath it. Right: everything else lives behind the menu button.

11. Definitions

Every term used in this guide, in plain English.

AGL

Above Ground Level. How high the antenna sits above the dirt directly beneath it. What you set on a mast, and what drone rules are written in.

ASL	Above Sea Level. Height measured from the sea. An aircraft holding a constant ASL altitude is at a changing AGL height as the ground rises and falls beneath it.
Airtime	The share of the channel your traffic occupies. Two radios cannot talk at the same instant, so everything - both directions, every stream - takes turns. Past roughly 75% the link feels broken even though the signal is strong.
Bandwidth (channel width)	How wide a slice of spectrum the radio uses, in MHz. Wider carries more data; narrower reaches further. It is a range control as much as a speed control.
BDA	Bi-Directional Amplifier. An external booster that raises both transmit power and receive sensitivity.
dB	Decibel - a ratio, not an amount. Every 3 dB doubles power; every 6 dB roughly doubles distance. A 10 dB difference is large.
dBi	Antenna gain. An antenna adds no power; it concentrates what it has into a narrower beam, so more gain means more reach but a tighter aim.
dBm	An actual power level. 30 dBm is 1 watt. Received signals are negative numbers: -60 dBm is strong, -90 dBm is nearly nothing.
Diffraction loss	Signal lost bending over a ridge or building between the two ends. The planner reads this from real elevation data.
Fade margin	Spare signal held back for bad days - rain, movement, reflections. Doodle Labs plan to 10 dB. A link with no margin works in the demonstration and fails in service.
Fresnel zone	The cigar-shaped volume around the straight line between two antennas. It needs to be mostly clear, so scraping a treeline costs signal even when you can see the far end.
Link margin	How much signal you have above the minimum the radio needs. Positive is good; the fade margin is subtracted from it.
MCS	Modulation and Coding Scheme - the data rate step the radio picks, 0 to 15. Higher is faster but needs a stronger signal, and the radio steps down automatically as conditions worsen.
Mesh	Radios that relay for each other, so traffic can route around a blocked path rather than simply failing.
Noise floor	The background hiss at the receiver. A noisy site needs a stronger signal to hear the same message, so a busy industrial yard costs real range.
Path loss	Everything the signal loses between the two antennas - distance, terrain, obstructions.
RSSI	Received Signal Strength Indicator. What the far radio actually hears, in dBm.

TDD / half duplex	The radio transmits and receives in turns on one channel. Uplink and downlink share the same airtime rather than having separate budgets.
Throughput	Data actually delivered, always less than the headline PHY rate once overheads, acknowledgements and retries are counted.
TX power	How hard the radio transmits, in dBm.

12. Where the numbers come from

Worth knowing, so you can judge how much to trust any given answer.

- **Radio performance** comes from Doodle Labs' own published figures - the transmit power and sensitivity for each model and data rate.
- **Terrain** comes from roughly 30-metre elevation data. It models the shape of the ground: hills and valleys, not buildings or trees.
- **Path loss** is free-space loss plus terrain diffraction, with an extra correction available for antennas close to the ground - which is where simple calculations are most optimistic.
- **Antenna patterns** are modelled from the beamwidths you enter, including the loss when a link sits off the centre of the beam.
- **Airtime and traffic** use a model designed by Aaron Do, Doodle Labs Solutions Engineering.

Be careful where it is optimistic

Antennas near the ground, links through buildings or forest, and busy spectrum are the three places predictions flatter reality. Real deployments have repeatedly measured a fraction of what a simple free-space calculation promised in exactly these conditions - which is why the near-ground correction and the noise floor setting exist. Use them, and keep your fade margin honest.

Questions, or something that looks wrong? Talk to Solutions Engineering - a case where the tool disagrees with the field is worth more to us than one where it agrees. Live tool: <https://doodlesim.jus2419497.workers.dev>.